LECTURE – DATA TYPES & VARIABLES OOP (JAVA)

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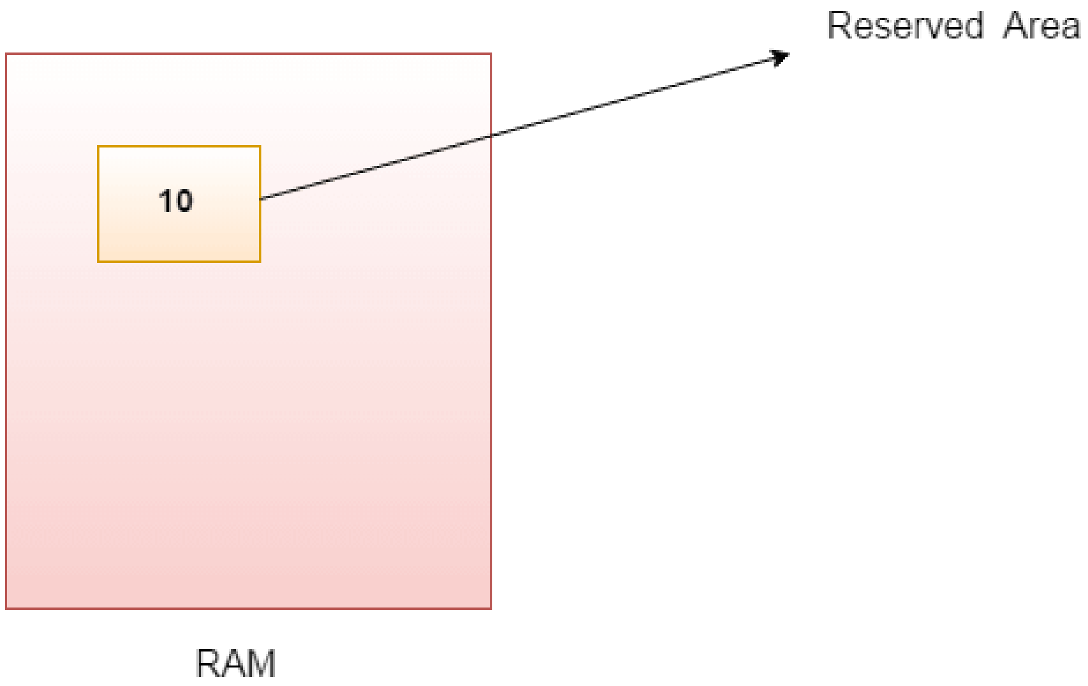
AGENDA

- Variables
- Types of Variables
  - Local
  - Class
  - Static
- Naming Rules
- Data Types
- Flavors of Data Types

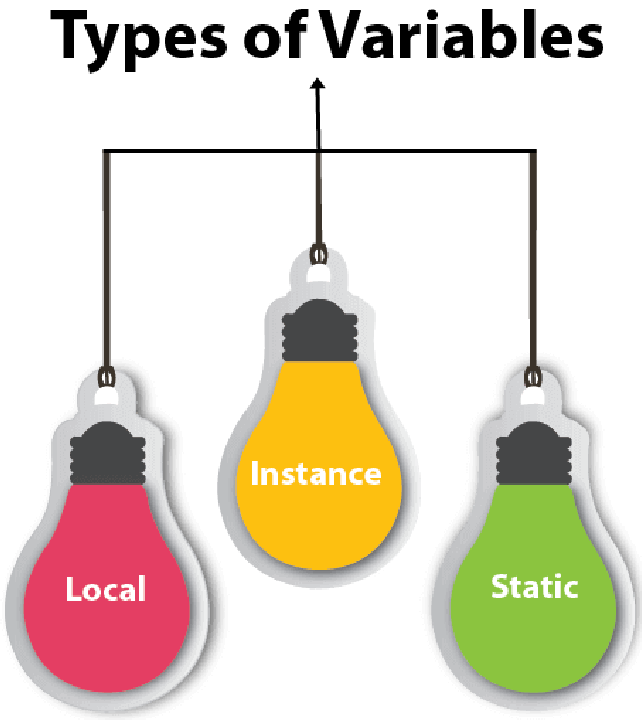
VARIABLES IN JAVA

- Variables are nothing but reserved memory locations to store values.
- This means that when you create a variable you reserve some space in memory.
- It is a combination of "vary + able" which means its value can be changed.

```
int data=50;//Here data is variable
```



VARIABLES IN JAVA



- There are three types of variables in Java:
  1. local variable
  2. instance variable
  3. static variable

LOCAL VARIABLES

- A variable declared inside the body of the method is called local variable.
- A local variable cannot be defined with "static" keyword.

  For example:

  int count = 0;

- As such, local variables are only visible to the methods in which they are declared.
- They are not accessible from the rest of the class.

INSTANCE VARIABLES (NON-STATIC FIELDS)

- A variable declared inside the class but outside the body of the method, is called an instance variable.
- It is not declared as static.
- It is called an instance variable because its value is instance-specific and is not shared among instances.
- Objects store their individual states in "non-static fields", that is, fields declared without the static keyword.
- Non-static fields are also known as instance variables because their values are unique to each instance of a class (to each object, in other words);
- For Example:
  - The currentSpeed of one bicycle is independent from the currentSpeed of another.

CLASS VARIABLES (STATIC FIELDS)

- A variable that is declared as static is called a static variable.
- It cannot be local.
- You can create a single copy of the static variable and share it among all the instances of the class.
- Memory allocation for static variables happens only once when the class is loaded in the memory.
- This tells the compiler that there is exactly one copy of this variable in existence, regardless of how many times the class has been instantiated.

The code:

```
static int numGears = 6;
```


would create such a static field.

PARAMETERS

- You've already seen examples of parameters, in the main method of the "Hello World!" application.
- Recall that the signature for the main method is:
  - public static void main(String[] args).
  - Here, the args variable is the parameter to this method.
- The important thing to remember is that parameters are always classified as "variables" not "fields".

NAMING CONVENTION FOR VARIABLES

- Variable names are case-sensitive.
- A variable's name can begin with a letter, the dollar sign "$", or the underscore character "_".
- The convention, however, is to always begin your variable names with a letter, not "$" or "_".
- White space between characters is not permitted.
- Also keep in mind that the name you choose must not be a keyword or reserved word.

DATA TYPES

- Data types specify the different sizes and values that can be stored in the variable.
- Based on the data type of a variable, the operating system allocates memory and decides what can be stored in the reserved memory.
- Therefore, by assigning different data types to variables, you can store integers, decimals, or characters in these variables.

DATA TYPES

- There are two data types available in Java:

1) Primitive Data Types

2) Reference/Object Data Types/Non-Primitive Data Types

PRIMITIVE DATA TYPES

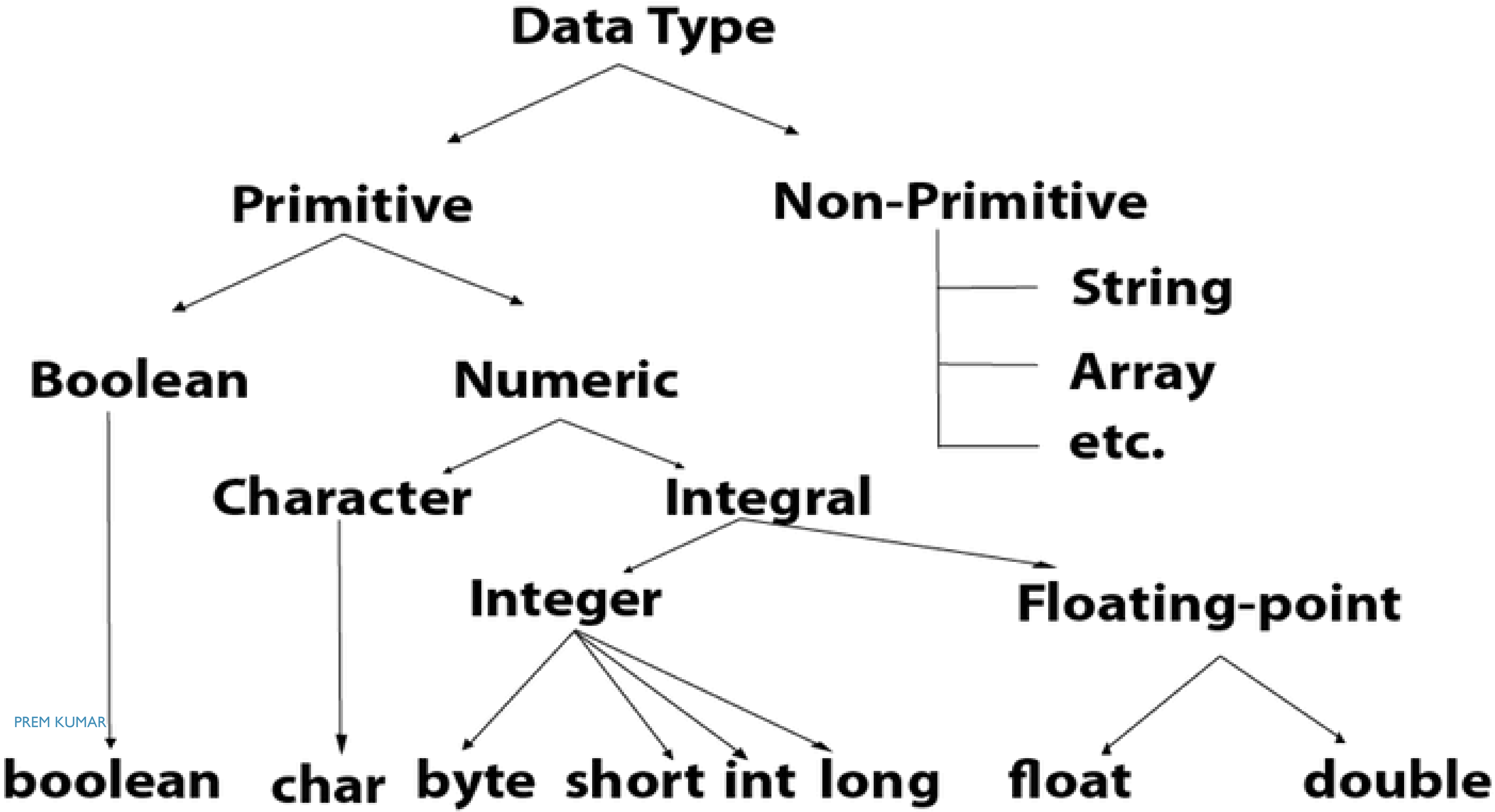
There are eight primitive data types supported by Java.

1. byte
2. short
3. int
4. long
5. float
6. double
7. char
8. boolean

In Java language, primitive data types are the building blocks of data manipulation. These are the most basic data types available in Java language.

Type	Contains	Default	Size	Range
`boolean`	`true` or `false`	`false`	1 bit	NA
`char`	Unicode character	`\u0000`	16 bits	`\u0000` to `\uFFFF`
`byte`	Signed integer	0	8 bits	-128 to 127
`short`	Signed integer	0	16 bits	-32768 to 32767
`int`	Signed integer	0	32 bits	-2147483648 to 2147483647
`long`	Signed integer	0	64 bits	-9223372036854775808 to 9223372036854775807
`float`	IEEE 754 floating point	0.0	32 bits	±1.4E-45 to ±3.4028235E+38
`double`	IEEE 754 floating point	0.0	64 bits	±4.9E-324 to ±1.7976931348623157E+308

PRIMITIVE DATA TYPES


Data Type
Primitive
Non-Primitive
String
Array
etc.
Boolean
Numeric
Character
Integral
Integer
Floating-point
boolean
char
byte
short
int
long
float
double

BYTE

- Byte data type is a 8-bit signed two's complement integer.
- Minimum value is -128 (-2^7)
- Maximum value is 127 (inclusive)(2^7 -1)
- Default value is 0
- Byte data type is used to save space in large arrays, mainly in place of integers.
- Example : byte a = 100 , byte b = -50

SHORT

- Short data type is a 16-bit signed two's complement integer.
- Minimum value is -32,768 (-2^15)
- Maximum value is 32,767(inclusive) (2^15 -1)
- Short data type can also be used to save memory as byte data type.
- Default value is 0.
- Example : short s= 10000 , short r = -20000

INT

- Int data type is a 32-bit signed two's complement integer.
- Minimum value is - 2,147,483,648.(-2^31)
- Maximum value is 2,147,483,647(inclusive).(2^31 -1)
- Int is generally used as the default data type for integral values unless there is a concern about memory.
- The default value is 0.
- Example : int a = 100000, int b = -200000

LONG

- Long data type is a 64-bit signed two's complement integer.
- Minimum value is -9,223,372,036,854,775,808.(-2^63)
- Maximum value is 9,223,372,036,854,775,807 (inclusive). (2^63 -1)
- This type is used when a wider range than int is needed.
- Default value is 0L.
- Example : int a = 100000L, int b = -200000L

FLOAT

- Float data type is a single-precision 32-bit IEEE 754 floating point.
- Float is mainly used to save memory in large arrays of floating point numbers.
- Default value is 0.0f.
- Float data type is never used for precise values such as currency.
- Example : float f1 = 234.5f

DOUBLE

- double data type is a double-precision 64-bit IEEE 754 floating point.
- This data type is generally used as the default data type for decimal values.
- Double data type should never be used for precise values such as currency.
- Default value is 0.0d.
- Example : double d1 = 123.4

BOOLEAN

- boolean data type represents one bit of information.
- There are only two possible values : true and false.
- This data type is used for simple flags that track true/false conditions.
- Default value is false.
- Example : boolean one = true

CHAR

- char data type is a single 16-bit Unicode character.
- Minimum value is '\u0000' (or 0).
- Maximum value is '\uffff' (or 65,535 inclusive).
- Char data type is used to store any character.
- Example . char letterA ='A'

JAVA ESCAPE CHARACTERS

Escape Sequence	Character Value
\b	Backspace
\t	Horizontal tab
\n	Newline
\f	Form feed
\r	Carriage return
\"	Double quote
\'	Single quote
\\	Backslash

REFERENCE/OBJECT DATA TYPES/ NON-PRIMITIVE DATA TYPES

There are an infinite number of possible class and array data types.

Collectively, these data types are known as *reference types*.

Class and array types differ significantly from primitive types, in that they are compound, or composite, types.

A primitive data type holds exactly one value.

Classes and arrays are aggregate types that contain multiple values.

JAVA LITERALS

- A literal is a source code representation of a fixed value.
- They are represented directly in the code without any computation.
- Literals can be assigned to any primitive type variable.
- For example:
  - byte a = 68;
  - char a = 'A'

Assignment:

Go and explore about Unicode System in Java with examples

Thanks ☺